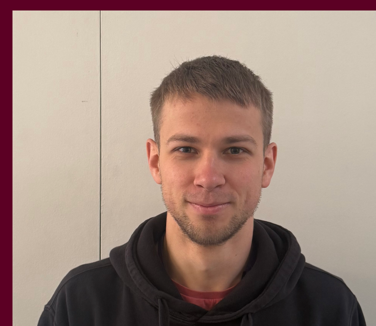
Theoretical, Quantum, and Computational Chemistry Research Talks – (Feb 17, 2026)

Multiscale Quantum-Classical Simulations of Organic Exciton-Polaritons: Dynamics, Delocalisation and Transport

Abstract

Experiments suggest that the intrinsic properties of organic photoactive materials can change when placed inside an optical microcavity, including their electrical and energy conductivity, photoemission quantum yield, or even chemical reactivity [1]. While these changes have been attributed to the strong coupling and hybridization between excitations of the material and the confined light modes of the cavity into polaritons, the current theoretical understanding of these phenomena is still very limited. To bridge this gap, a method combining established models from quantum optics with established models from computational chemistry, namely Quantum Mechanics/Molecular Mechanics (QM/MM), was developed [2]. The method allows to perform atomistic simulations of the dynamics of large molecular ensembles with more than a thousand molecules coupled to multiple confined light modes of optical cavities. In my talk, the main ideas behind the method will be discussed, and several applications will be provided, including how strong coupling influences transport of excitons [3], polariton relaxation [4], photochemistry [5], and exciton-exciton annihilation.



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Dr. Ilia was born in Petropavlovsk-Kamchatsky, Russia. He completed both his BSc (Electronics and Nanoelectronics) and MSc (Solar Heterostructure Photovoltaic Energetics) at Saint Petersburg Electrotechnical University "LETI". He then earned his PhD in Chemistry at University of Jyväskylä, Finland; his doctoral research focused on multiscale molecular dynamics simulations of enhanced excitation energy transport in organic microcavities. He is currently a Royal Society International Newton Research Fellow at University College London, United Kingdom.

References:

[1] R. Bhuyan et al. *Chem. Rev.* 2023, 123, 18, 10877–10919
[2] H.L. Luk et al. *J. Chem. Theory Comput.* 2017, 13, 9, 4324–4335
[3] I. Sokolovskii et al. *Nat. Commun.* 2023, 14, 6613
[4] R.H. Tichauer et al. *J. Phys. Chem. Lett.* 2022, 13, 27, 6259–6267
[5] A. Dutta et al. *Nat. Commun.* 2023, 15, 6600

HOW TO CONNECT

TUESDAY, Feb 17, 2026
5:00 P.M. - 6:00 P.M. (CST)
11:00 P.M. - 12:00 A.M. (GMT)

Video Conference Details:

Join Zoom Meeting
https://us04web.zoom.us/j/73193181498?pwd=ydlHAFmbomxJReG1ILNpxHSsAcvTWN.1

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